Dual-Sensor Vibration Analysis System

(Transfer Function & Resonance Measurement)

Overview

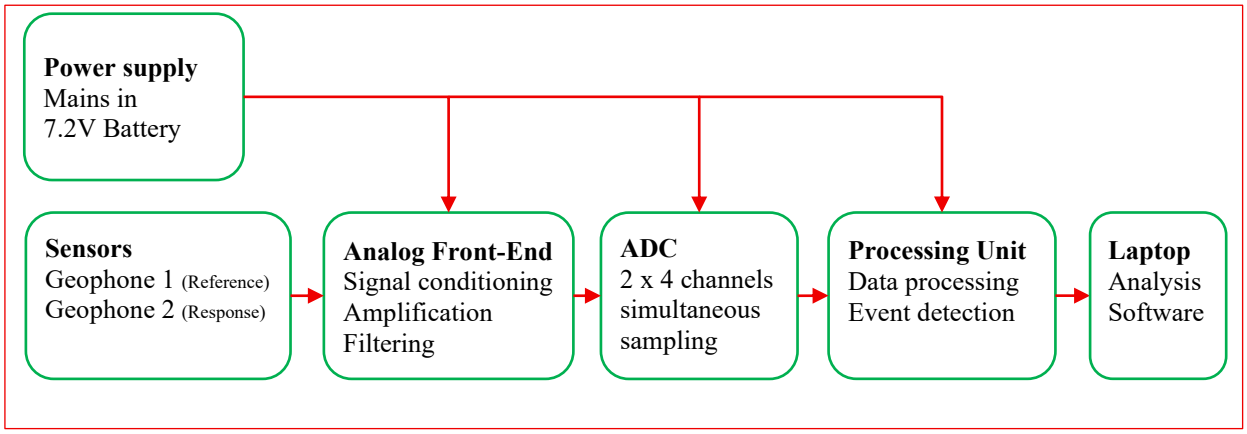
A compact, dual-channel measurement system designed for vibration analysis, isolation characterization, and structural resonance detection. The system simultaneously acquires signals from two sensors, enabling direct comparison between reference and response measurements.

It is optimized for laboratory and field testing of vibration isolation platforms, structural dynamics, and mechanical systems. The system is designed for direct data acquisition and transfer to PC, enabling real-time visualization, storage, and detailed post-processing such as transfer function and frequency response analysis.

Key Capabilities

- Dual-sensor synchronous acquisition:
  - Simultaneous sampling of reference and response signals
  - Phase-aligned measurement for transfer function computation
- Vibration isolation characterization:
  - Measures attenuation between ground and isolated platform
  - Enables evaluation of isolation effectiveness across frequency range
- Structural resonance detection:
  - Identifies natural frequencies and resonance peaks
  - Supports structural analysis and validation
- High-resolution data acquisition:
  - 16-bit SAR ADC with multi-channel support
  - Wide dynamic range for low and high amplitude signals
- PC-based analysis:
  - Real-time data streaming to PC/laptop
  - Supports both live monitoring and offline analysis
- Power system:
  - 7.2V battery (~10Ah) with mains input

System Architecture



Typical Measurement Outputs

Transfer Function

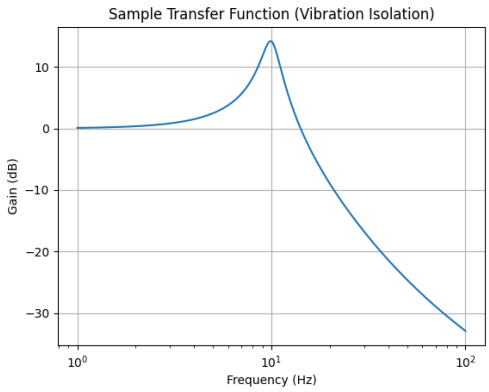


Figure 1: Transfer function showing vibration attenuation between input (ground) and output (isolated platform)

Resonance Curve

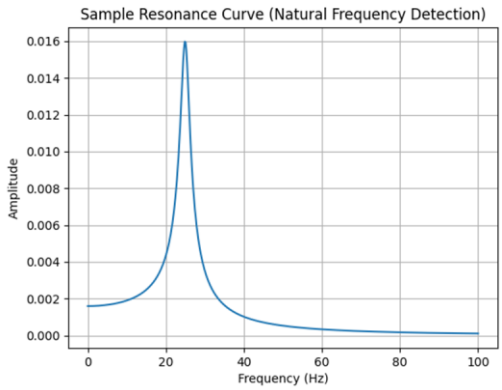


Figure 2 Frequency response showing resonance peak used to identify natural frequency

Typical Applications

- Vibration Isolation Characterization
  - Evaluation of isolation platforms and damping systems
  - Transfer function measurement between input and output
- Structural Resonance / Natural Frequency Detection
  - Identification of structural resonance points
  - Validation of mechanical and structural designs
- Laboratory Vibration Analysis
  - Controlled testing of vibration-sensitive equipment
  - Performance verification of isolation tables